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Prometheus

Automated crypto trading — the desktop engine, in the browser. Generates its own signals, auto-executes trades and manages risk 24/7.

Full-stack automated crypto trading platform · FastAPI + ccxt · v2.0 · July 26, 2026

1 · What is Prometheus?

Prometheus is a multi-user, non-custodial crypto trading platform. Each customer connects their own exchange with a **trade-only API key** (withdrawals disabled), and the bot runs a fully-automated strategy 24/7 on their account — generating its own buy/sell signals, executing entries and exits, and managing risk. One FastAPI server hosts many users; funds never leave the user's exchange.

```
Browser dashboard ↙
TradingView webhook + → FastAPI → per-user TraderSession → ccxt → exchange
Built-in strategy ↘      (guardrails → sizing → order → SL/TP bracket)
```

Key features

Fully-automated strategy

EMA 9/21 crossover + 100-EMA trend filter, confirmation candles, swing-based stops, partial take-profit + trailing exit. ON by default; fully tunable.

Live signal-strength meter

A 0–100% gauge shows the next trade forming; at 85% it fires a Telegram + push "get ready" alert seconds before entry.

High-impact news filter

One-tap switch to hold new entries 1h before & after major USD events (FOMC, CPI, jobs). Open trades stay managed.

Weekend pause

Optionally hold new entries from Saturday until Sunday evening, then auto-resume for the market re-open.

Scale-in (pyramiding)

Adds a smaller same-direction trade once 40% to TP and moves the original stop to break-even.

Binance Binance.US Bybit OKX KuCoin Bitget Kraken Coinbase — spot & futures

Profit Simulator

Replays the exact live strategy on real market history — win rate, drawdown, equity curve, every trade.

Telegram — two-way control

Entry/close alerts + daily PnL recap, and run the bot from chat: `/status /pnl /positions /pause /resume /closeall`.

VIP Custom Strategy

Advanced users unlock every parameter — a one-time **\$497** for lifetime access, granted via an invoice flow by the admin.

Security

Fernet-encrypted trade-only keys, 2FA (TOTP), login history, sign-out-all, withdrawal-key warning, IP-whitelist helper.

Diagnostics

Public `/api/health` + authenticated `/api/selftest` (DB, exchange ping, market feed, strategy loop, licence) with a one-click Settings button.

2 · Project structure & stack

Path	What it is
server/main.py	FastAPI app — REST + WebSocket routes, auth, admin, webhooks, health/selftest
server/session.py	Per-user TraderSession — strategy loop, order flow, live state, weekend/news guards
server/store.py	SQLite (WAL) data layer — users, keys, trades, licences, audit
server/security.py	Password hashing, JWT, Fernet key encryption, TOTP 2FA
server/engine/	Trading core — exchange.py (ccxt), strategy.py, guardrails.py, backtest.py
web/	Dashboard front-end — index.html, app.js, admin.js, style.css (vanilla JS + TradingView Lightweight Charts)
Dockerfile · fly.toml	Container + Fly.io deploy config

Stack: Python 3.11 · FastAPI · ccxt · SQLite · Fernet · PyJWT · WebSockets · vanilla JS.

3 · Run locally

```
cd app.prometheus
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
export PROMETHEUS_SECRET_KEY="$(python -c 'import secrets;print(secrets.token_urlsafe(48))')"
uvicorn server.main:app --host 0.0.0.0 --port 8000
# open http://localhost:8000 (first registered user becomes admin)
```

Environment variables

Var	Purpose
PROMETHEUS_SECRET_KEY	Required. Long random string — encrypts API keys & signs JWTs. Protect with a secret manager.
PROMETHEUS_ADMIN_EMAIL	Email that becomes admin on registration.
PROMETHEUS_PUBLIC_URL	Your public base URL (e.g. https://yourdomain.com) — used in reset links & webhooks.
PROMETHEUS_DATA_DIR	Where the SQLite DB lives — point at a persistent path so data survives restarts.
PROMETHEUS_TRIAL_DAYS	Free-trial length (default 10).

4 · Move to any server — one-click install

Prometheus is a standard FastAPI service, so it runs on any VPS with SSH + Python — **Hostinger VPS**, **GoDaddy VPS**, **DigitalOcean**, **Contabo** and the like. A bundled installer does the entire setup for you.

One command

```
# upload (or git clone) the app folder to the server, then:
cd app.prometheus
sudo bash install.sh
```

It asks for your **domain** and **admin email**, then automatically:

- installs Python, Nginx & all dependencies
- creates a virtualenv and installs `requirements.txt`
- generates & saves a strong `PROMETHEUS_SECRET_KEY` into `/etc/prometheus.env`
- installs a **systemd service** — auto-starts on boot, auto-restarts on crash
- configures an **Nginx reverse proxy** with WebSocket support
- issues a **free auto-renewing HTTPS certificate** (Let's Encrypt)

Then open `https://yourdomain.com` — the first account you register becomes the admin. Manage with `systemctl status|restart|stop prometheus`; to update, re-upload files (or `git pull`) then `systemctl restart prometheus`. Re-running `install.sh` is safe — it keeps your existing secret key & database.

No VPS? Hostinger / cPanel shared hosting

On cPanel → "Setup Python App": set the app root to the uploaded folder, startup file `passenger_wsgi.py`, Python 3.11, install `requirements.txt`, and add the env vars. Passenger doesn't proxy WebSockets, so the dashboard auto-falls back to 3-second polling (still fully functional). For true real-time updates use a VPS with the installer above.

Quick alternative — Fly.io (containers)

```
cd app.prometheus
fly deploy
fly secrets set PROMETHEUS_SECRET_KEY=... PROMETHEUS_ADMIN_EMAIL=...
```

GitHub Actions also auto-deploys on push to `main`. The marketing site (`site.prometheus/`) deploys separately to Hostinger from the `site.prometheus` branch. **Full cPanel steps on the next page.**

Appendix · cPanel / shared hosting — step by step

No VPS or SSH? Any cPanel host with "Setup Python App" (Hostinger shared, GoDaddy cPanel, Namecheap...) runs Prometheus through Passenger. WebSockets aren't proxied, so the dashboard auto-uses 3-second polling — everything still works.

1 · Upload the app

cPanel → **File Manager** → your home folder (keep it *outside* `public_html`). Upload the zip and **Extract** it — e.g. `/home/USER/prometheus` .

2 · Create the Python app

cPanel → **Setup Python App** → **Create Application:**

Field	Value
Python version	3.11 (or the highest available)
Application root	<code>prometheus</code> (the folder you extracted)
Application URL	your domain or subdomain
Application startup file	<code>passenger_wsgi.py</code>
Application Entry point	<code>application</code>

Click **Create**.

3 · Install dependencies

On the app's page, under **"Configuration files"** add `requirements.txt` and click **Run Pip Install** — or copy the shown `source .../activate` command into cPanel **Terminal** and run `pip install -r requirements.txt` .

4 · Set environment variables

Same page, under **Environment variables**, add:

```
PROMETHEUS_SECRET_KEY = <a long random string, 40+ chars>
PROMETHEUS_PUBLIC_URL  = https://yourdomain.com
PROMETHEUS_DATA_DIR    = /home/USER/prometheus/data
PROMETHEUS_ADMIN_EMAIL = you@email.com
```

Need a secret? Any strong random-string generator works, or run `python -c "import secrets;print(secrets.token_urlsafe(48))"` on any PC.

5 · Restart & open

Click **Restart** on the app page, then open your domain — the **first account you register becomes the admin**.

Updating later: re-upload the changed files in File Manager, then click **Restart**. **Real-time:** Passenger uses 3-second polling instead of live WebSockets — for instant updates use the one-click VPS installer instead.

5 · Admin panel (/admin)

- Every customer: status, keys, last-seen, PnL — **suspend/activate, grant/revoke licences** (N days or lifetime).
- **Custom-strategy (VIP) requests**: mark an invoice paid to unlock a customer's custom strategy (\$497).
- Global managed strategy (pushed to all managed users), broadcasts, social-proof, audit log.
- **One-click database backup & restore** — export the whole DB (customers, encrypted keys, trades, licences) as a file and import it on another server. Keep the same `PROMETHEUS_SECRET_KEY` so encrypted keys still decrypt.
- First registered user is admin; or set `PROMETHEUS_ADMIN_EMAIL`.

6 · VIP Custom Strategy — invoice flow

1. Admin grants a **lifetime licence** → the customer sees the ★ Unlock VIP offer.
2. Customer taps **Request access** → shows in the admin requests panel.
3. Admin emails an **invoice for \$497**; on payment clicks **Mark invoice paid**.
4. Custom Strategy unlocks instantly; the customer's top bar shows the ★ **VIP Unlocked** badge.

⚠ **Trading risk & custody.** Users' exchange keys are stored encrypted server-side and the server trades on their behalf — the operator is a custodian. Protect `PROMETHEUS_SECRET_KEY` with a real KMS, require trade-only keys (withdrawals off), and keep users in paper/safe mode until validated. Provided as-is, no warranty. Trading involves risk and can lose money.